Mise en place d’un service WEB

Mettre en place une machine virtuelle Rocky Linux

Mettre à jour le gestionnaire de paquet :

```
[root@localhost ~]# sudo dnf update -y
sudo dnf upgrade -y
Rocky Linux 10 - BaseOS                                          11 MB/s |  14 MB     00:01
Rocky Linux 10 - AppStream                                      5.4 MB/s | 2.3 MB     00:00
Rocky Linux 10 - Extras                                          21 kB/s | 6.0 kB     00:00
Dépendances résolues.
```

Installation du pare-feu et mise en place de celui-ci :

```
[root@localhost ~]# sudo dnf install -y vim wget curl firewalld
En attente de la fin d’exécution du processus ayant l’identifiant (pid) 33045.
Le paquet curl-8.12.1-2.el10_1.2.x86_64 est déjà installé.
Le paquet firewalld-2.4.0-1.el10_1.noarch est déjà installé.
Dépendances résolues.
```

```
[root@localhost ~]# ^C
[root@localhost ~]# sudo systemctl enable firewalld
[root@localhost ~]# sudo systemctl start firewalld
[root@localhost ~]#
[root@localhost ~]# sudo firewall-cmd --permanent --add-service=http
success
[root@localhost ~]# sudo firewall-cmd --permanent --add-service=https
success
[root@localhost ~]#
[root@localhost ~]# sudo firewall-cmd --reload
success
[root@localhost ~]#
```

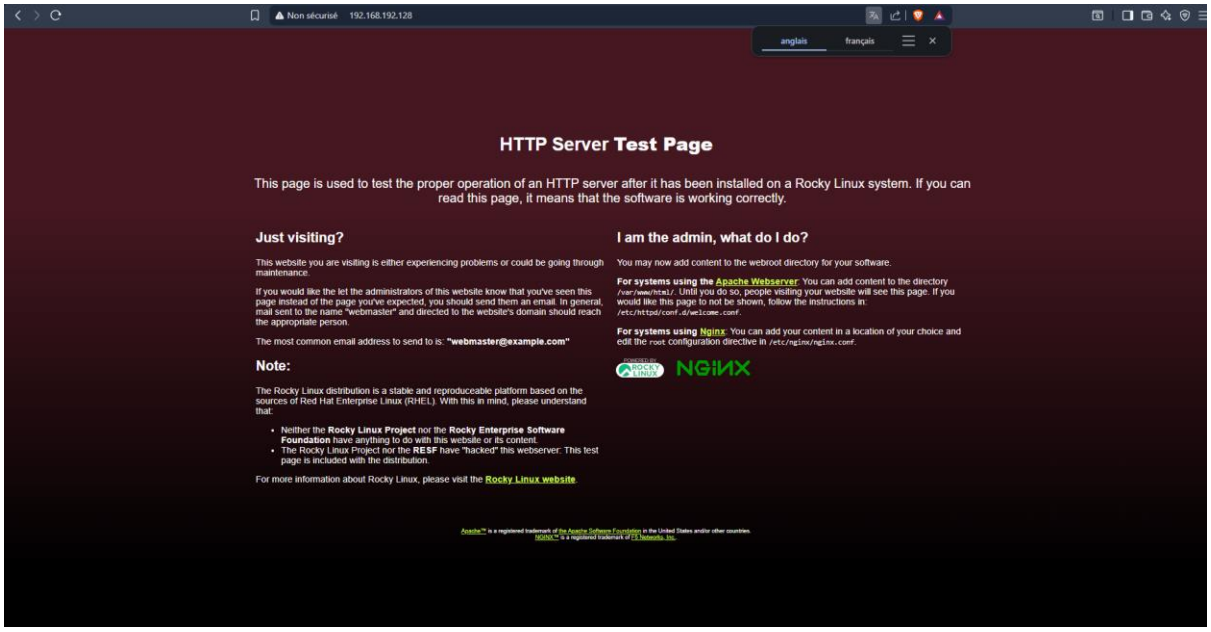
Installation et mise en place de nginx

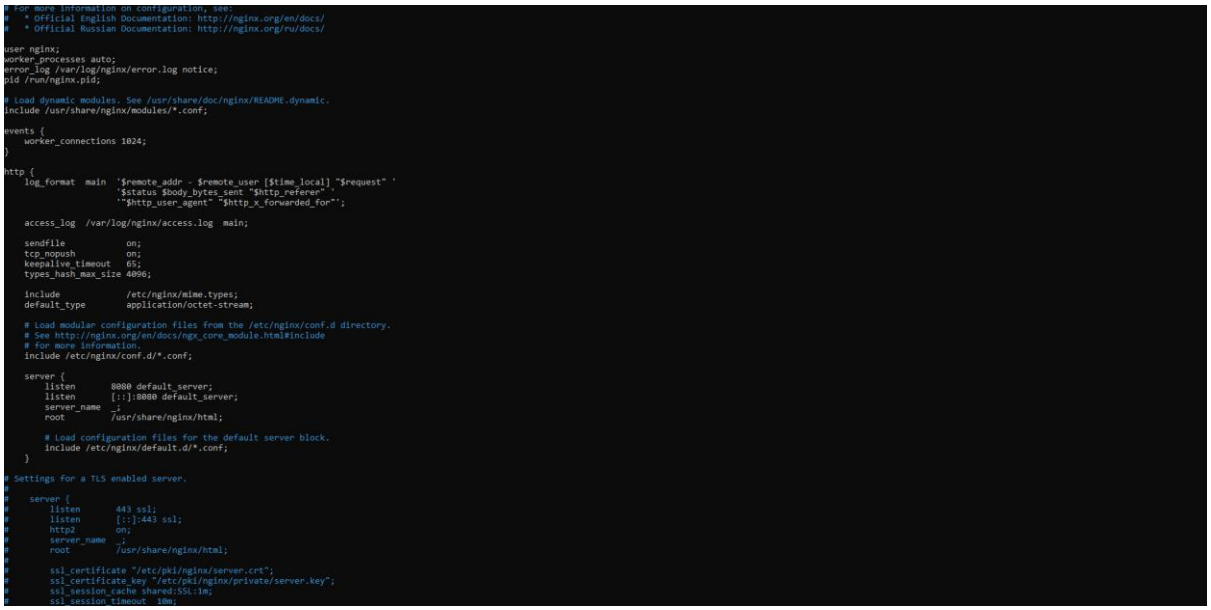
```
[root@localhost ~]# sudo dnf install -y nginx
sudo systemctl enable nginx
sudo systemctl start nginx
Dernière vérification de l’expiration des métadonnées effectuée il y a 0:04:22 le dim. 22 mars 2026 15:13:40.
Dépendances résolues.
```

Installation et mise en place de mariadb

```
[root@localhost ~]# sudo dnf install -y mariadb-server
sudo systemctl enable mariadb
sudo systemctl start mariadb
```

On peut se connecter sur le site web en mettant l’IP :





```
# For more information on configuration, see:
#   * Official English Documentation: http://nginx.org/en/docs/
#   * Official Russian Documentation: http://nginx.org/ru/docs/

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
                      '$status $body_bytes_sent "$http_referer" '
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/ngx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    server {
        listen       8080 default_server;
        listen       [::]:8080 default_server;
        server_name  _;
        root         /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;
    }

# Settings for a TLS enabled server.
#
#    server {
#        listen       443 ssl;
#        listen       [::]:443 ssl;
#        http2        on;
#        server_name  _;
#        root         /usr/share/nginx/html;
#
#        ssl_certificate "/etc/pki/nginx/server.crt";
#        ssl_certificate_key "/etc/pki/nginx/private/server.key";
#        ssl_session_cache shared:SSL:1m;
#        ssl_session_timeout  10m;
```

Création de la page index du site web

```
[root@localhost www]# mkdir -p web
[root@localhost www]# ls
cgi-bin  html  web
[root@localhost www]# cd web/
[root@localhost web]#
```

```
  GNU nano 8.1                                    index.html
Bonjour je suis le serveur de Tokyo
```

Changement de configuration

```
  GNU nano 8.1                                      web.conf
server {
 listen 80;
 listen [::]:80;
 root /var/www/web/ ;
 index index.html index.htm index.nginx-debian.html sign-up.php;
 server_name web.fr ;
 location ~* \.php$ {
 fastcgi_pass unix:/run/php-fpm/www.sock;
 include fastcgi_params;
 fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
 fastcgi_param SCRIPT_NAME $fastcgi_script_name;
}
 access_log /var/log/nginx/access_web.log;
 error_log /var/log/nginx/error_web.log;
 location / {
 try_files $uri $uri/ =404;
 }
}
```

```
[root@localhost /]# mkdir /etc/ssl/private
[root@localhost /]# chmod 700 /etc/ssl/private
```

Création de l’auto certif

```
[root@localhost /]# openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/nginx
selfsigned.key -out /etc/ssl/certs/nginx-selfsigned.cr
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+........+.......................+.+..+.+......+.....+....+.........+.....++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:FR
State or Province Name (full name) []:
Locality Name (eg, city) [Default City]:
Organization Name (eg, company) [Default Company Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) []:
Email Address []:
[root@localhost /]#
```

```
[root@localhost /]# openssl dhparam -out /etc/ssl/certs/dhparam.pem 2048
Generating DH parameters, 2048 bit long safe prime
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[root@localhost /]#
```

```
  GNU nano 8.1                                    web.conf
server {
 listen 80;
 listen [::]:80;
 root /var/www/web/ ;
 index index.html index.htm index.nginx-debian.html sign-up.php;
 server_name web.fr ;
 location ~*
\.php$ {
 fastcgi_pass unix:/run/php
-fpm/www.sock;
 include fastcgi_params;
 fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
 fastcgi_param SCRIPT_NAME $fastcgi_script_name;
}
 access_log /var/log/nginx/access_web.log;
 error_log /var/log/nginx/error_web.log;
 location / {
 try_files $uri $uri/ =404;
 }
}
server {
 listen 443 http2 ssl;
 listen [::]:443 http2 ssl;
ssl_certificate /etc/ssl/certs/nginx-selfsigned.crt;
 ssl_certificate_key /etc/ssl/private/nginx
-selfsigned.key;
 ssl_dhparam /etc/ssl/certs/dhparam.pem;
 root /var/www/web/ ;
 index index.html index.htm index.nginx
-debian.html sign
-up.php;
 server_name web.fr ;
 location ~*
\.php$ {
 fastcgi_pass unix:/run/php
-fpm/www.sock;
 include fastcgi_params;
 fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
 fastcgi_param SCRIPT_NAME $fastcgi_script_name;
}
 access_log /var/log/nginx/access_web.log;
 error_log /var/log/nginx/error_web.log;
 location / {
 try_files $uri $uri/ =404;
}
}
```

```
[root@localhost conf.d]#  nginx -t
nginx: [warn] the "listen ... http2" directive is deprecated, use the "http2" directive instead in /etc/nginx/conf.d/web.conf:20
nginx: [warn] the "listen ... http2" directive is deprecated, use the "http2" directive instead in /etc/nginx/conf.d/web.conf:21
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[root@localhost conf.d]#
```

```
[root@localhost /]# systemctl restart nginx
[root@localhost /]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
    Drop-In: /etc/systemd/system/nginx.service.d
             └─php-fpm.conf
     Active: active (running) since Mon 2026-03-23 09:11:17 CET; 17s ago
 Invocation: 141e11dcc9ae4c25905c79c66d09d5e1
    Process: 5255 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 5257 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 5259 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 5260 (nginx)
      Tasks: 5 (limit: 10374)
     Memory: 5M (peak: 5.5M)
        CPU: 40ms
     CGroup: /system.slice/nginx.service
             ├─5260 "nginx: master process /usr/sbin/nginx"
             ├─5261 "nginx: worker process"
             ├─5262 "nginx: worker process"
             ├─5263 "nginx: worker process"
             └─5264 "nginx: worker process"

mars 23 09:11:17 localhost.localdomain systemd[1]: Starting nginx.service - The nginx HTTP and reverse p
mars 23 09:11:17 localhost.localdomain nginx[5257]: nginx: [warn] the "listen ... http2" directive is de
mars 23 09:11:17 localhost.localdomain nginx[5257]: nginx: [warn] the "listen ... http2" directive is de
mars 23 09:11:17 localhost.localdomain nginx[5257]: nginx: the configuration file /etc/nginx/nginx.conf 
mars 23 09:11:17 localhost.localdomain nginx[5257]: nginx: configuration file /etc/nginx/nginx.conf test
mars 23 09:11:17 localhost.localdomain nginx[5259]: nginx: [warn] the "listen ... http2" directive is de
mars 23 09:11:17 localhost.localdomain nginx[5259]: nginx: [warn] the "listen ... http2" directive is de
mars 23 09:11:17 localhost.localdomain systemd[1]: Started nginx.service - The nginx HTTP and reverse pr
[root@localhost /]#
```

```
[root@localhost ~]# dnf install haproxy -y
Rocky Linux 10 - BaseOS                                          14 MB/s |  14 MB     00:01
Rocky Linux 10 - AppStream                                      3.6 MB/s | 2.3 MB     00:00
Rocky Linux 10 - Extras                                          15 kB/s | 6.0 kB     00:00
Dépendances résolues.
================================================================================================
 Paquet                  Architecture           Version                          Dépôt          Taille
================================================================================================
Installation:
 haproxy                 x86_64                 3.0.5-4.el10_1.1                 appstream        2.6 M

Résumé de la transaction
================================================================================================
Installer  1 Paquet

Taille totale des téléchargements : 2.6 M
Taille des paquets installés : 8.0 M
Téléchargement des paquets :
haproxy-3.0.5-4.el10_1.1.x86_64.rpm                              8.4 MB/s | 2.6 MB     00:00
------------------------------------------------------------------------------------------------
Total                                                            5.0 MB/s | 2.6 MB     00:00
Rocky Linux 10 - AppStream                                       1.6 MB/s | 1.6 kB     00:00
Import de la clef GPG 0x6FEDFC85 :
Utilisateur : « Release Engineering (Rocky Linux 10) <releng@rockylinux.org> »
```

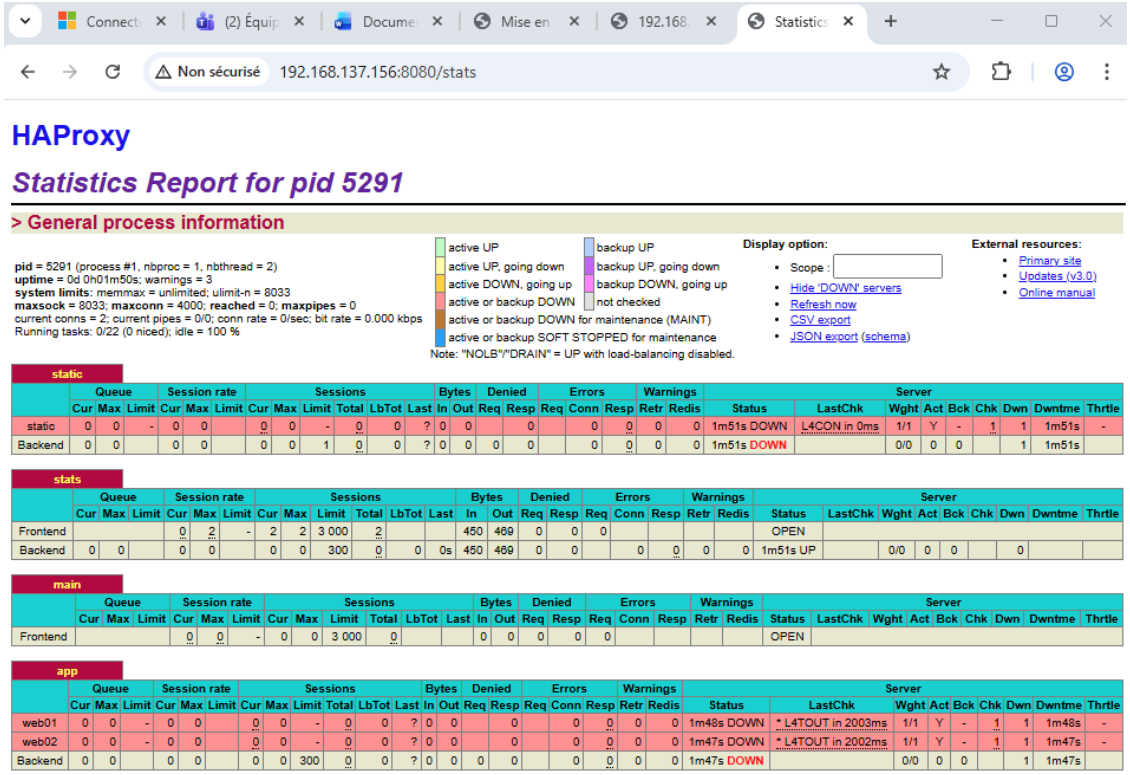
```
#---------------------------------------------------------------------
# round robin balancing between the various backends
#---------------------------------------------------------------------
backend app
    balance     roundrobin
    server  app1 127.0.0.1:5001 check
    server  app2 127.0.0.1:5002 check
    server  app3 127.0.0.1:5003 check
    server  app4 127.0.0.1:5004 check

listen stats
 bind *:8080
 mode http
 stats enable
 stats hide-version
 stats uri /stats
 stats admin if LOCALHOST
 stats auth haproxy:haproxy

#---------------------------------------------------------------------
# main frontend which proxys to the backends
#---------------------------------------------------------------------
frontend main
 bind *:80
 acl url_static path_beg -i /static /images /javascript /stylesheets
 acl url_static path_end -i .jpg .gif .png .css .js
# use_backend static if url_static
default_backend app

#---------------------------------------------------------------------
# round robin balancing between the various backends
#---------------------------------------------------------------------
backend app
 balance roundrobin
 server web01 192.168.1.39:443 ssl verify none check
 server web02 192.168.1.49:443 ssl verify none check
```

```
# rsyslog configuration file

# For more information see /usr/share/doc/rsyslog-*/rsyslog_conf.html
# or latest version online at http://www.rsyslog.com/doc/rsyslog_conf.html
# If you experience problems, see http://www.rsyslog.com/doc/troubleshoot.html

#### GLOBAL DIRECTIVES ####

# Where to place auxiliary files
global(workDirectory="/var/lib/rsyslog")

#### MODULES ####

# Use default timestamp format
module(load="builtin:omfile" Template="RSYSLOG_TraditionalFileFormat")

module(load="imuxsock"    # provides support for local system logging (e.g. via logger command)
       SysSock.Use="off") # Turn off message reception via local log socket;
                          # local messages are retrieved through imjournal now.
module(load="imjournal"             # provides access to the systemd journal
       UsePid="system" # PID nummber is retrieved as the ID of the process the journal entry originat>
       FileCreateMode="0644" # Set the access permissions for the state file
       StateFile="imjournal.state") # File to store the position in the journal

# Include all config files in /etc/rsyslog.d/
include(file="/etc/rsyslog.d/*.conf" mode="optional")

#module(load="imklog") # reads kernel messages (the same are read from journald)
#module(load="immark") # provides --MARK-- message capability

# Provides UDP syslog reception
# for parameters see http://www.rsyslog.com/doc/imudp.html
module(load="imudp") # needs to be done just once
input(type="imudp" port="514")

# Provides TCP syslog reception
# for parameters see http://www.rsyslog.com/doc/imtcp.html
#module(load="imtcp") # needs to be done just once
#input(type="imtcp" port="514")

#### RULES ####

# Log all kernel messages to the console.
# Logging much else clutters up the screen.
#kern.*                                                 /dev/console

# Log anything (except mail) of level info or higher.
```

```
[root@localhost haproxy]# systemctl restart haproxy
[root@localhost haproxy]# systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
     Loaded: loaded (/usr/lib/systemd/system/haproxy.service; disabled; preset: disabled)
     Active: active (running) since Mon 2026-03-23 10:52:42 CET; 8s ago
 Invocation: a1111777c3254b02ab9b3402b6149cac
    Process: 5141 ExecStartPre=/usr/sbin/haproxy -f $CONFIG -f $CFGDIR -c -q $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 5144 (haproxy)
     Status: "Ready."
      Tasks: 3 (limit: 10364)
     Memory: 8M (peak: 9.6M)
        CPU: 36ms
     CGroup: /system.slice/haproxy.service
             ├─5144 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.pid
             └─5146 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.pid
```

```
[root@localhost haproxy]# systemctl start rsyslog
[root@localhost haproxy]# systemctl status rsyslog
● rsyslog.service - System Logging Service
     Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-23 09:44:23 CET; 1h 10min ago
 Invocation: 9c8a40e9058a49f2bbae688aeada76fe
       Docs: man:rsyslogd(8)
             https://www.rsyslog.com/doc/
   Main PID: 1034 (rsyslogd)
      Tasks: 3 (limit: 10364)
     Memory: 2.9M (peak: 3.5M)
        CPU: 423ms
     CGroup: /system.slice/rsyslog.service
             └─1034 /usr/sbin/rsyslogd -n
```

Interface web :




```
[root@localhost certs]# openssl req -x509 -nodes -newkey rsa:2048 -keyout /etc/pki/tls/certs/haproxy.p
em -out /etc/pki/tls/certs/haproxy.pem -days 365
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-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:FR
```

Mise en place du load balancer

```
#---------------------------------------------------------------------
# round robin balancing between the various backends
#---------------------------------------------------------------------
backend app
 balance roundrobin
 server web01 192.168.137.153:443 ssl verify none check
 server web02 192.168.137.157:443 ssl verify none check
```